

A Thesis Defense Presentation

ASYMMETRIC J-CURVE IN NEPAL'S PEGGED EXCHANGE RATE REGIME

In Partial Fulfilment of the Requirements for the Degree of Master of Arts in Economics

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BACKGROUND OF THE STUDY

- **The Theoretical Expectation:**
 - Currency depreciation → exports cheaper, imports costlier → trade balance should improve (Marshall–Lerner condition)
 - **J-Curve:** trade balance worsens first, improves later (Magee, 1973 — contract lags, adjustment delays)
- **Nepal's Unique Setting:**
 - Dual exchange rate regime — Fixed peg: NPR ↔ INR (since 1993); Float: NPR vs. USD and others
 - REER moves only via INR cross-rates & inflation differentials
- **The Paradox:**
 - NPR depreciated 20 years vs. convertible currencies, yet trade deficit widened persistently → contradicts conventional theory

STATEMENT OF THE PROBLEM

- **Theory vs. Reality:**
 - J-Curve predicts: worsens first → improves later
 - Nepal: 20 years depreciation → deficit widens
- **Central Question:**
 - Why does devaluation fail in Nepal?
- **Possible Explanations:**
 - Model? (symmetry assumption)
 - Structure? (inelastic trade)
 - Regime? (peg vs. float)

OBJECTIVES OF THE RESEARCH

1. To examine the nature of the asymmetric J-Curve effect in Nepal's aggregate trade balance
2. To assess the asymmetric response to real appreciation versus real depreciation in the long run
3. To analyze the asymmetric J-curve effects between Nepal's trade with India (pegged regime) and the Rest of the World (floating regime) form 1980 to 2024.

RESEARCH HYPOTHESES

- **H1 — Asymmetric J-Curve in Aggregate Trade**

- H_0 : No asymmetric J-Curve effect in Nepal's aggregate trade balance
- H_1 : An asymmetric J-Curve effect exists

- **H2 — Asymmetric Long-Run Adjustment**

- H_0 : Trade balance responds symmetrically to appreciation and depreciation ($\theta^+ = \theta^-$)
- H_1 : Responses are asymmetric — Wald test rejects $\theta^+ = \theta^-$ at the 5% level

- **H3 — Regime-Dependent Asymmetry**

- H_0 : Asymmetric response is identical for India (pegged) and ROW (floating) trade
- H_1 : The asymmetric response differs between regimes

LITERATURE REVIEW & RESEARCH GAP

- **1. THEORY — What Should Happen**

- Marshall–Lerner: devaluation improves TB if $\eta_X + \eta_M > 1$
- Magee (1973): J-Curve — contract lags cause initial deterioration before improvement
- Goldstein & Khan (1985): $TB = f(Y, Y^*, REER)$

- **2. METHODS — How Testing Evolved**

- Linear era → mixed, inconclusive results (Bahmani-Oskooee 1985; Rose & Yellen 1989)
- Asymmetric turn → Bahmani-Oskooee & Fariditavana (2015/16): J-curves HIDDEN under linear models emerge once symmetry is relaxed
- Shin et al. (2014) NARDL → now the standard for asymmetry testing (Arize et al. 2017; Bhat & Bhat 2021)

- **3. NEPAL — What's Been Done**

- Chaulagai (2015); Pathak (2020); Adhikari (2020); Bishwakarma et al. (2024)
- All LINEAR models → mixed result

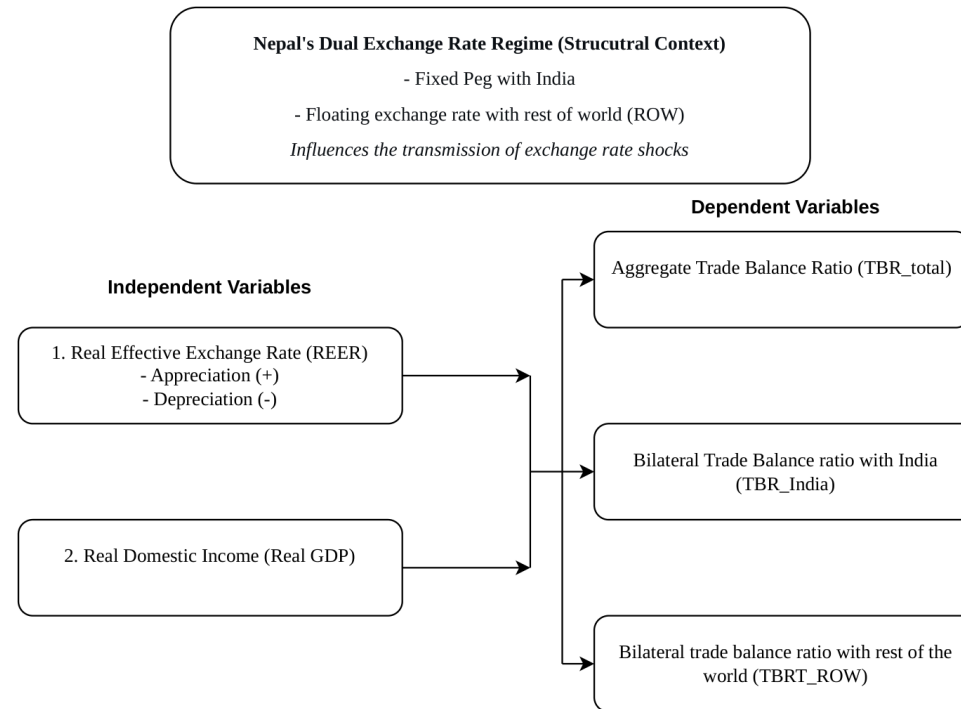
- **THE GAP:**

- No NARDL study exists for Nepal

RESEARCH DESIGN

- **Nature:** Quantitative, time-series study
- **Data:** Annual data, 1980–2024 (45 observations)
- **Variables:** REER (split into appreciation / depreciation) + Real GDP and Trade Balance
- **Models:** Estimated separately for Aggregate, India (pegged), ROW (floating)
- **Software Used:** Eviews 12

CONCEPTUAL FRAMEWORK



DATA AND VARIABLES

Variable	Definition	Measurement	Source
ln TBR (total, India, ROW)	Trade Balance Ratio = Exports / Imports	Natural log	NRB Quarterly Economic Bulletin
ln REER	Real Effective Exchange Rate	Natural log, index	Bruegel Database ¹
REER⁺ / REER⁻	Partial sums of REER	Appreciation / depreciation components	Author's construction
ln Nepal GDP	Real domestic income	Natural log	World Bank WDI database

- Annual data, 1980–2024 (45 observations), estimated in EViews 12

¹Bruegel datasets, <https://www.bruegel.org/publications/datasets/real-effective-exchange-rates-for-178-countries-a-new-database>

ESTIMATION STRATEGY & MODEL SPECIFICATION

- **STAGE 1 – LINEAR ARDL (Pesaran et al., 2001):** baseline + cointegration

$$\Delta \text{TBR}_t = \alpha_0 + \sum \alpha_i \Delta \text{TBR}_{t-i} + \sum \beta_j \Delta \text{REER}_{t-j} + \sum \rho_v \Delta \text{GDP}_{t-v} + \delta_1 \text{TBR}_{t-1} + \delta_2 \text{REER}_{t-1} + \delta_3 \text{GDP}_{t-1} + \varepsilon_t$$

- Bounds F-test; assumes SYMMETRY by construction
- **STAGE 2 – ECM:** ECT = speed of adjustment back to equilibrium (sign, size, significance)

$$\Delta \text{TBR}_t = \alpha_0 + \sum \alpha_i \Delta \text{TBR}_{t-i} + \sum \beta_j \Delta \text{REER}_{t-j} + \sum \rho_v \Delta \text{GDP}_{t-v} + \lambda \text{ECT}_{t-1} + \varepsilon_t$$

- **STAGE 3 – NARDL (Shin et al., 2014):** asymmetry + Wald test

$$\Delta \text{TBR}_t = \alpha_0 + \sum \alpha_i \Delta \text{TBR}_{t-i} + \sum \beta_j^+ \Delta \text{REER}_{t-j}^+ + \beta_j^- \Delta \text{REER}_{t-j}^- + \sum \rho_v \Delta \text{GDP}_{t-v} + \lambda \text{ECT}_{t-1} + \varepsilon_t$$

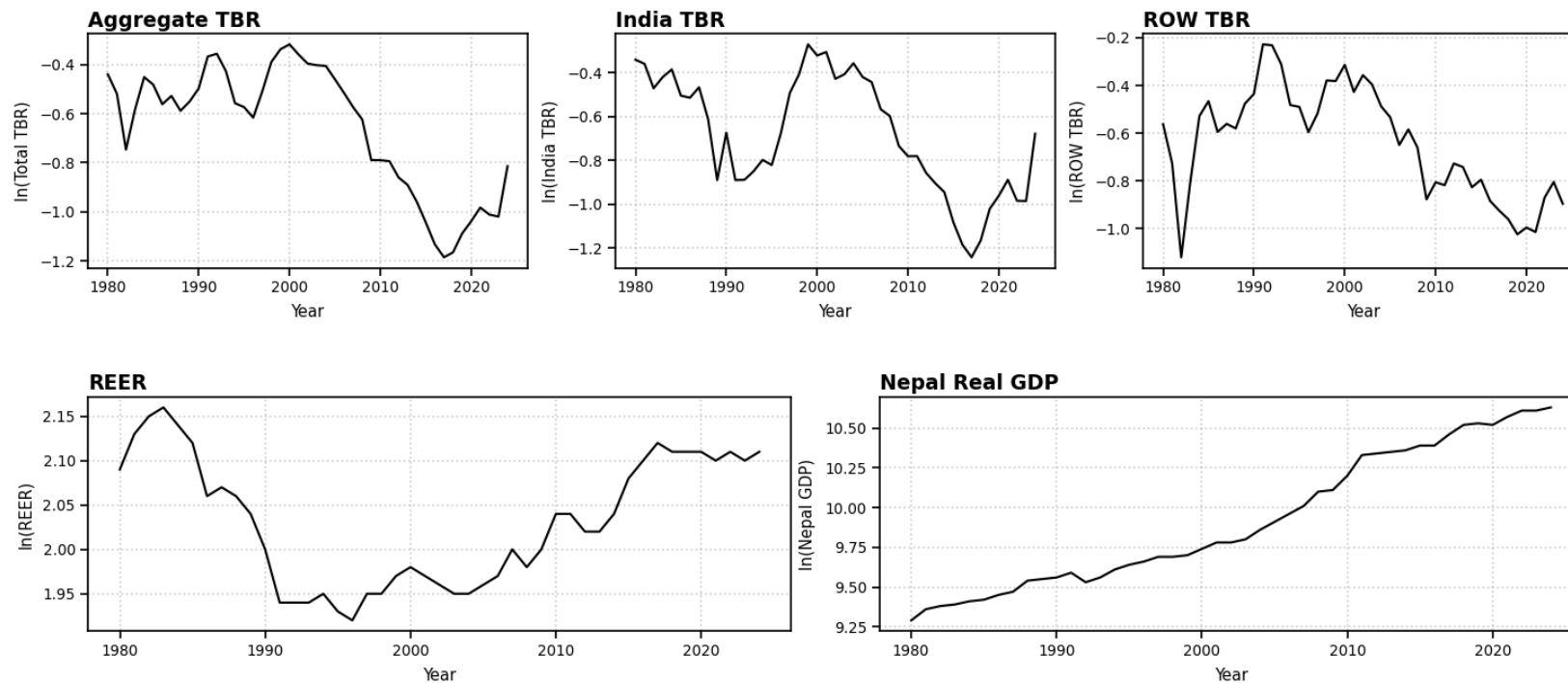
- REER^+ = appreciation partial sums; REER^- = depreciation partial sums
- **Wald:** $H_0 : \theta^+ = \theta^-$ vs. $H_1 : \theta^+ \neq \theta^-$; lags by AIC / SIC
- **Logic:** SYMMETRIC world $\rightarrow$ ASYMMETRIC world

DESCRIPTIVE STATISTICS (1980–2024, N = 45)

Variable	Mean	SD	Min	Median	Max	Skew.	Kurt.	JB	JB p
ln total tbr	−0.66	0.26	−1.18	−0.57	−0.32	−0.59	−0.98	4.42	0.11
ln india tbr	−0.68	0.27	−1.24	−0.68	−0.27	−0.24	−1.09	2.68	0.26
ln row tbr	−0.64	0.23	−1.12	−0.59	−0.23	−0.14	−0.96	1.89	0.39
ln reer	2.03	0.07	1.92	2.04	2.16	0.10	−1.42	3.86	0.14
ln nepal gdp	9.92	0.43	9.29	9.78	10.63	0.32	−1.36	4.21	0.12
ln india gdp	11.86	0.43	11.27	11.71	12.59	0.25	−1.46	4.45	0.11
ln row gdp	14.48	0.33	13.94	14.41	14.96	−0.17	−1.24	3.09	0.21

- **KEY TAKEAWAYS:**
 - All variables pass Jarque-Bera normality ($p > 0.05$) → supports parametric estimation

DATA TRENDS (1980–2024)



PRE-ESTIMATION DIAGNOSTICS: UNIT ROOT TESTS

- **Purpose:** order of integration — **Tests:** ADF and PP, at level and first difference

Variable	ADF (Level)	PP (Level)	ADF (1st Diff.)	PP (1st Diff.)	Decision
ln total tbr	-1.448 (0.549)	-1.151 (0.686)	-4.476 (0.000)***	-4.321 (0.001)***	I(1)
ln reer	-1.620 (0.463)	-1.141 (0.690)	-4.817 (0.000)***	-4.813 (0.000)***	I(1)
ln nepal gdp	0.386 (0.980)	0.379 (0.979)	-5.917 (0.000)***	-5.904 (0.000)***	I(1)
ln india tbr	-1.690 (0.429)	-1.884 (0.336)	-5.593 (0.000)***	-5.723 (0.000)***	I(1)
ln row tbr	-1.662 (0.443)	-1.585 (0.481)	-6.121 (0.000)***	-6.503 (0.000)***	I(1)

Note. Parentheses = *p*-values. *** = significant at 1%.

- **CONCLUSION:** all variables I(1), none I(2) → ARDL / NARDL bounds testing is VALID

STAGE 1 RESULTS: LINEAR ARDL BOUNDS TEST, LONG-RUN COEFFICIENTS AND MODEL FIT

Model	ARDL	F-stat	I(0)	I(1)	Cointegration?	R ²	Adj. R ²
Aggregate	(2, 0, 0)	2.009	3.100	3.870	No	0.919	0.911
India	(1, 0, 0)	0.962	3.100	3.870	No	0.836	0.824
ROW	(2, 0, 0)	5.731	3.100	3.870	Yes	0.826	0.808

- Long-run relationships exists in ROW model only
- **ROW long-run (only cointegrated model):**

Variable	Coef.	Std. Error	t-Stat.	Prob.
ln reer	-1.767	0.291	-6.059	0.000***
ln nepal gdp	-0.281	0.049	-5.632	0.000***
C	5.745	0.609	9.421	0.000***

STAGE 2 RESULTS: ERROR CORRECTION MODEL (ECM)

- **ECM equation (fitted values, ROW):**

$$\Delta \text{TBR}_t = \alpha_0 + \sum \gamma_i \Delta \text{TBR}_{t-i} + (-1.484)^{**} \cdot \Delta \text{REER}_t + \sum \psi_i \Delta \text{GDP}_{t-i} + (-0.839)^{***} \cdot \text{ECT}_{t-1} + \varepsilon_t$$

Model	SR REER	Prob.	ECT(-1)	t-stat.	Prob.
Aggregate	-0.382	0.163	-0.335	-2.685	0.010***
India	-0.145	0.557	-0.150	-1.755	0.086*
ROW	-1.484	0.002***	-0.839	-4.572	0.000***

Note. ECT = Error Correction Term. * $p < .10$, ** $p < .05$, *** $p < .01$.

- **Reading:** ROW ECT = -0.839^{***} → 83.9% of disequilibrium corrected per period (fast adjustment)
- Aggregate/India ECT weakly significant despite no cointegration – low-power artifact ($T = 43$; Kremers et al., 1992); J-curve reading applies to ROW only

STAGE 3 RESULTS: NARDL ESTIMATION

Model	Order	F-stat	5% Bounds	Cointegration?	R ²	Adj. R ²
Aggregate	(2, 0, 0, 0)	1.711	[2.79, 3.67]	No	0.92	0.91
India	(1, 0, 0, 0)	0.979	[2.79, 3.67]	No	0.841	0.824
ROW	(3, 0, 0, 0)	8.318	[2.79, 3.67]	Yes	0.898	0.881

Regime	Variable	Coef.	Std. Err.	t-stat.	Prob.	Sig.
Aggregate	REER POS	0.666	2.588	0.257	0.798	
	REER NEG	-1.541	0.765	-2.014	0.051	*
	GDP	-0.989	0.772	-1.280	0.208	
India	REER POS	6.749	9.391	0.718	0.476	
	REER NEG	-2.656	2.912	-0.911	0.367	
	GDP	-2.606	2.867	-0.909	0.368	
ROW	REER POS	0.245	0.849	0.288	0.774	
	REER NEG	-1.800	0.331	-5.436	0.000	***
	GDP	-0.848	0.257	-3.299	0.002	***

STAGE 3 (cont'd): NARDL SHORT-RUN & ERROR CORRECTION

Model	SR REER POS	Prob.	SR REER NEG	Prob.	ECT(-1)	t-stat	Prob.
Aggregate	0.218	0.791	-0.505	0.115	-0.328	-2.602	0.013**
India	0.941	0.394	-0.370	0.269	-0.139	-1.613	0.115
ROW	0.227	0.772	-1.668	0.006***	-0.927	-4.787	0.000***

ASYMMETRY TESTING: WALD TEST (H2)

- Long-run restriction $\theta^+ = \theta^-$ tested at 5%

Model	Wald F	Prob.	Decision (5%)
Aggregate	0.604	0.441	Symmetric
India	1.041	0.313	Symmetric
ROW	3.467	0.071	Asymmetric (at 10%)

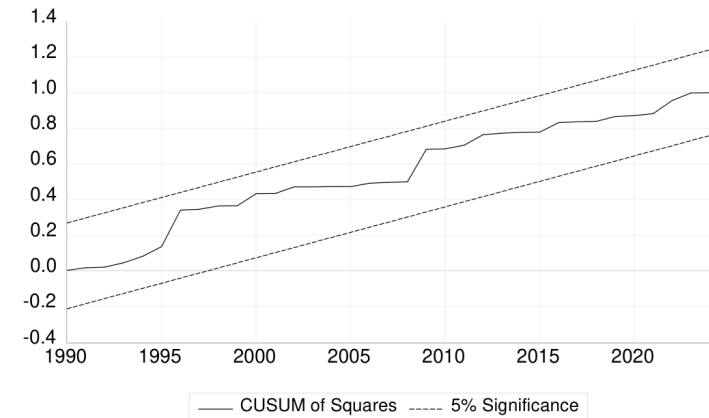
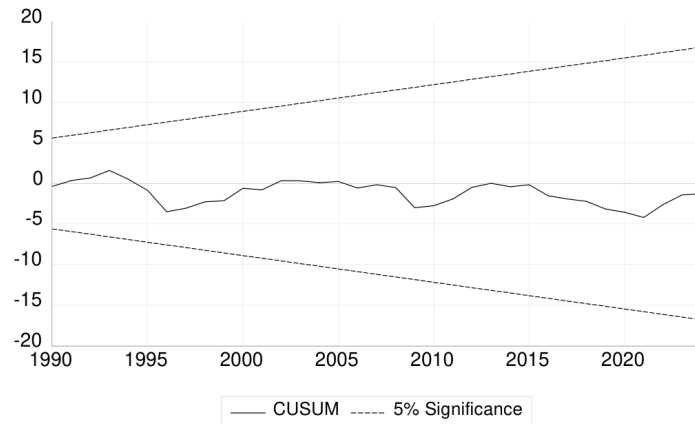
Note. H_0 = long-run symmetry of REER+ / REER- effects. Rejection = asymmetric long-run effects.

- **Reading:** symmetry holds for Aggregate/India; ROW rejects at 10% → long-run asymmetry lives in the floating regime only — H2 partially supported

DIAGNOSTIC TESTING: NARDL MODELS

- BG (serial corr.) + BPG (heterosked.) + Ramsey RESET (functional form)

Model	BG (p)	BPG (p)	RESET (p)
Aggregate	0.991	0.022**	0.515
India	0.883	0.736	0.586
ROW	0.730	0.509	0.644



HYPOTHESIS TESTING SUMMARY

Hypothesis	Verdict	Evidence
H1: Asymmetric J-Curve in aggregate trade	✗ REJECTED	No J-pattern anywhere; ROW shows deterioration WITHOUT recovery ($\theta^- = -1.800^{***}$)
H2: Asymmetric long-run adjustment ($\theta^+ \neq \theta^-$)	✓ SUPPORTED	Wald rejects symmetry for ROW at 10% ($F = 3.467$, $p = 0.071$); Aggregate/India symmetric
H3: Regime-dependent asymmetry (India $\neq$ ROW)	✓ SUPPORTED	India: no cointegration, symmetric (peg mutes transmission); ROW: cointegration + asymmetry; NARDL-ECM ECT = -0.926 (faster than linear -0.839)

DISCUSSION — WHY DOES DEPRECIATION FAIL NEPAL?

- **1. Depreciation improves ROW Trade:**
 - Matches the region: India (Bhat & Bhat, 2021); US (Nasir & Leung, 2021)
 - Marshall–Lerner holds — asymmetrically
 - Explains Nepal's linear null result - the improvement was hidden in the ROW segment (Chaulagai, 2025; Bishwakarma et al., 2024: linear models found no J curve)
- **2. Appreciation does nothing: One way adjustment**
 - Import demand saturated — cheaper imports don't raise volumes (Pathak, 2020; Bahmani-Oskooee & Hajilee, 2009)
 - Down work, up doesn't
- **3. Growth is the deficit engine's:**
 - Every 1% GDP growth → 0.85% TBR deterioration
 - Remittances ≈ 25% of GDP fuel import demand (NRB)
 - Growth worsens the balance faster than depreciation improves it
- **4. Peg Blocks the working channel:**
 - Study confirms the Adhikari (2020); Nepal India transmission has no cointegration and no channel
 - Further, NPR's REER follows INR

CONCLUSIONS & POLICY RECOMMENDATIONS

CONCLUSIONS

1. J-Curve **REJECTED** — no initial deterioration; improvement is immediate
2. Marshall-Lerner **HOLDS** - asymmetrically: depreciation improves ROW trade balance; appreciation does not
3. The peg neutralizes transmission entirely for the India trade
4. The deficit's engine is import intensive growth
5. So, depreciation works but in ONE direction, in ONE regime, on a lever that Nepal does not control.
6. Exchange rate policy alone cannot fix the deficit.

POLICY RECOMMENDATIONS

1. **Export diversification & capacity building** — trade responds to capacity, not price
2. **Inflation control (near India's)** — under the peg, inflation is the only real-rate channel
3. **Productive use of remittances** — into investment, not import consumption
4. **Structural reforms** — cut import dependence; build domestic production

QUESTIONS?

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